



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Practical No.: 5

Aim –

Use JavaScript function types, scope, closures; apply try-catch; build a palindrome checker.

Software/Tools Required –

VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

Experiment Program Code –

Index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Shopping Cart Calculator</title>

  <style>
    body {
      font-family: Arial;
      background: #f2f2f2;
      margin: 0;
      padding: 0;
    }

    .container {
      width: 800px;
      margin: 30px auto;
      background: white;
      padding: 20px;
      border-radius: 10px;
      box-shadow: 0 0 10px gray;
    }

    h2 {
      text-align: center;
    }

    input {
      padding: 8px;
      width: 150px;
      margin: 5px;
    }

    button {
      padding: 10px;
```

```

        cursor: pointer;
    }

    table {
        width: 100%;
        border-collapse: collapse;
        margin-top: 20px;
    }

    table, th, td {
        border: 1px solid black;
    }

    th {
        background: #007bff;
        color: white;
    }

    th, td {
        padding: 10px;
        text-align: center;
    }

    .result {
        margin-top: 20px;
        font-size: 18px;
    }

    footer {
        margin-top: 30px;
        padding: 15px;
        text-align: center;
        background: #007bff;
        color: white;
        border-radius: 5px;
        font-size: 15px;
    }
</style>
</head>

<body>

    <div class="container">

        <h2>Shopping Cart Calculator</h2>

        <input type="text" id="name" placeholder="Product Name">

        <input type="number" id="price" placeholder="Price">

        <input type="number" id="qty" placeholder="Quantity">

        <button onclick="addProduct()">Add Product</button>

        <table id="cartTable">
            <tr>
                <th>ID</th>
                <th>Product</th>
                <th>Price</th>
                <th>Quantity</th>
                <th>Total</th>
            </tr>
        </table>

        <div class="result" id="result">

            <h3>Item Summary</h3>

```

```

        <ul id="summary"></ul>

        <h3>Expensive Products (Price > 1000)</h3>
        <ul id="expensive"></ul>

    </div>

    <footer>
        Developed by Sanskruti Gorle | PRN - 24070521025
    </footer>

</div>

<script>

    let cart = [];

    function addProduct() {

        let name = document.getElementById("name").value;

        let price = parseFloat(
            document.getElementById("price").value
        );

        let qty = parseInt(
            document.getElementById("qty").value
        );

        if (name === "" || isNaN(price) || isNaN(qty)) {

            alert("Please enter all fields");

            return;
        }

        let product = {

            id: cart.length + 1,

            name: name,

            price: price,

            quantity: qty
        };

        cart.push(product);

        displayCart();

        document.getElementById("name").value = "";

        document.getElementById("price").value = "";

        document.getElementById("qty").value = "";
    }

    function displayCart() {

        let table = document.getElementById("cartTable");

        table.innerHTML = `
            <tr>
                <th>ID</th>
                <th>Product</th>
                <th>Price</th>
            </tr>
        `;
    }

```

```

        <th>Quantity</th>
        <th>Total</th>
    </tr>
`;

cart.forEach(function(item) {

    table.innerHTML += `
        <tr>
            <td>${item.id}</td>
            <td>${item.name}</td>
            <td>${item.price}</td>
            <td>${item.quantity}</td>
            <td>${item.price * item.quantity}</td>
        </tr>
    `;

});

let total = cart.reduce(function(sum, item) {

    return sum + (item.price * item.quantity);

}, 0);

let discount = 0;

if (total >= 50000) {

    discount = total * 0.20;

}

else if (total >= 20000) {

    discount = total * 0.10;

}

else if (total >= 5000) {

    discount = total * 0.05;

}

let finalAmount = total - discount;

document.getElementById("result").innerHTML = `

    <b>Total Amount :</b> ${total}<br>

    <b>Discount :</b> ${discount.toFixed(2)}<br>

    <b>Final Amount :</b> ${finalAmount.toFixed(2)}

    <h3>Item Summary</h3>

    <ul id="summary"></ul>

    <h3>Expensive Products (Price > 1000)</h3>

    <ul id="expensive"></ul>

`;

```

```

let summary = document.getElementById("summary");

summary.innerHTML = "";

cart.map(function(item) {

    summary.innerHTML +=
        `<li>${item.name} : ${item.price * item.quantity}</li>`;

});

let expensive = document.getElementById("expensive");

expensive.innerHTML = "";

let exp = cart.filter(function(item) {

    return item.price > 1000;

});

exp.forEach(function(item) {

    expensive.innerHTML +=
        `<li>${item.name}</li>`;

});

}

</script>
</body>
</html>

```

Output –

The screenshot shows a web browser window with the title "Shopping Cart Calculator". The address bar shows the file path: "C:/Users/SANSKRUTI/Downloads/javascript/practical%205/index.html". The application is displayed in a white box with a blue border. It features a form with input fields for "Product Name", "Price", and "Quantity", and an "Add Product" button. Below the form is a table with the following data:

ID	Product	Price	Quantity	Total
1	Laptop	70000	1	70000
2	Charger	3000	1	3000
3	Mouse	700	1	700

Below the table, the application displays the following summary:

Total Amount : 73700
Discount : 14740.00
Final Amount : 58960.00

Item Summary

- Laptop : 70000
- Charger : 3000
- Mouse : 700

Expensive Products (Price > 1000)

- Laptop
- Charger

At the bottom of the application box, a blue bar contains the text: "Developed by Sanskruti Corle | PRN - 24070521025". The browser's status bar at the bottom shows the temperature as 26°C, the language as ENG IN, and the date as 08-08-2026.

Case Study Title –

To study the functioning of Arrays and Loops in Java Script

a) Create and manipulate arrays: push, pop, shift, unshift, splice, slice. b) Use array methods: mapO, filter(), reduce(), forEach (.

Case Study: -

Create an array of numbers and find the maximum and minimum value.

Case Study Program Code –

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Maximum & Minimum Finder</title>

  <link rel="stylesheet" href="style.css">

  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>

<div class="container">

  <div class="icon"><img alt="Icon" data-bbox="114 558 138 572"/></div>

  <h1>Maximum & Minimum Finder</h1>

  <p class="subtitle">
    Enter array elements separated by commas
  </p>

  <label>Array Elements</label>

  <input
    type="text"
    id="numbers"
    placeholder="Example : 12,45,89,6,34">

  <button onclick="findValues()">
    Analyze Array
  </button>

  <div id="result"></div>

</div>

<script src="script.js"></script>

<footer class="footer">
  Designed & Developed by
  <strong>Sanskriti Gorle</strong><br>
  <span>PRN: 24070521025</span>
```

```
</footer>
</body>
</html>
```

style.css

```
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:'Poppins',sans-serif;
}

body{

  min-height:100vh;

  display:flex;
  flex-direction:column;
  justify-content:center;
  align-items:center;

  background:#F7F3EE;

  background-image:
    radial-gradient(circle at top left,#F3E5D7 0%,transparent 35%),
    radial-gradient(circle at bottom right,#E8DCCF 0%,transparent 35%);

  padding:20px;
}

.container{

  width:410px;

  background:#FFFDFC;

  padding:28px;

  border-radius:22px;

  border:1px solid #EFE6DD;

  box-shadow:0 15px 35px rgba(95,72,52,.12);
}

.icon{

  width:65px;
  height:65px;

  margin:auto;

  display:flex;
  justify-content:center;
  align-items:center;

  border-radius:50%;

  font-size:30px;

  background:#E8D8C7;
```

```
    color:#6B4F3B;
    margin-bottom:15px;
}

h1{
    text-align:center;
    color:#4B3621;
    font-size:24px;
    margin-bottom:5px;
    font-weight:700;
}

.subtitle{
    text-align:center;
    color:#8B735D;
    margin-bottom:20px;
    font-size:14px;
}

label{
    display:block;
    color:#5F4B3A;
    font-weight:500;
    margin-bottom:8px;
}

input{
    width:100%;
    padding:12px;
    border:2px solid #E6D7C8;
    border-radius:14px;
    outline:none;
    background:white;
    font-size:15px;
    transition:.3s;
}

input:placeholder{
    color:#B39B87;
}

input:focus{
    border-color:#A47551;
```

```
    box-shadow:0 0 0 4px rgba(164,117,81,.12);
}

button{

    width:100%;

    margin-top:16px;

    padding:12px;

    border:none;

    border-radius:14px;

    background:#8B5E3C;

    color:white;

    font-size:15px;

    font-weight:600;

    cursor:pointer;

    transition:.3s;
}

button:hover{

    background:#6F4A2F;

    transform:translateY(-2px);

    box-shadow:0 10px 18px rgba(111,74,47,.20);
}

button:active{

    transform:scale(.98);
}

#result{

    display:none;

    margin-top:18px;

    background:#FCFAF8;

    border:1px solid #EFE3D8;

    border-radius:16px;

    padding:15px;

    animation:fade .4s ease;
}

.result-title{

    font-size:17px;

    font-weight:600;

    color:#4B3621;
```

```
    margin-bottom:12px;
}

.row{

    display:flex;

    justify-content:space-between;

    align-items:center;

    padding:10px 12px;

    margin:8px 0;

    background:#F8F4EF;

    border-radius:10px;
}

.label{

    color:#6F5B4A;

    font-weight:500;
}

.value{

    font-weight:600;

    color:#4B3621;

    text-align:right;

    word-break:break-word;
}

.max {

    color:#2E7D32;
}

.min{

    color:#C0392B;
}

.footer{

    margin-top:18px;

    padding-top:14px;

    border-top:1px solid #EFE3D8;

    text-align:center;

    font-size:12px;

    color:#8B735D;

    line-height:20px;
}

.footer strong{
```

```
color:#5F4B3A;

font-weight:600;
}

.footer span{

color:#A08A76;

font-size:12px;
}

@keyframes fade{

from{

opacity:0;
transform:translateY(12px);

}

to{

opacity:1;
transform:translateY(0);

}

}
```

script.js

```
function findValues(){

let input = document.getElementById("numbers").value.trim();

if(input==""){

alert("Please enter array elements.");
return;

}

let arr = input.split(",").map(Number);

let max = arr[0];
let min = arr[0];

arr.forEach(function(number){

if(number > max){
max = number;
}

if(number < min){
min = number;
}

});

let result = document.getElementById("result");

result.style.display = "block";
```

```

result.innerHTML = `

<div class="result-title">
  <img alt="Clipboard icon" data-bbox="165 135 185 150"/> Analysis Result
</div>

<div class="row">
  <span class="label">Array</span>
  <span class="value">${arr.join(", ")}</span>
</div>

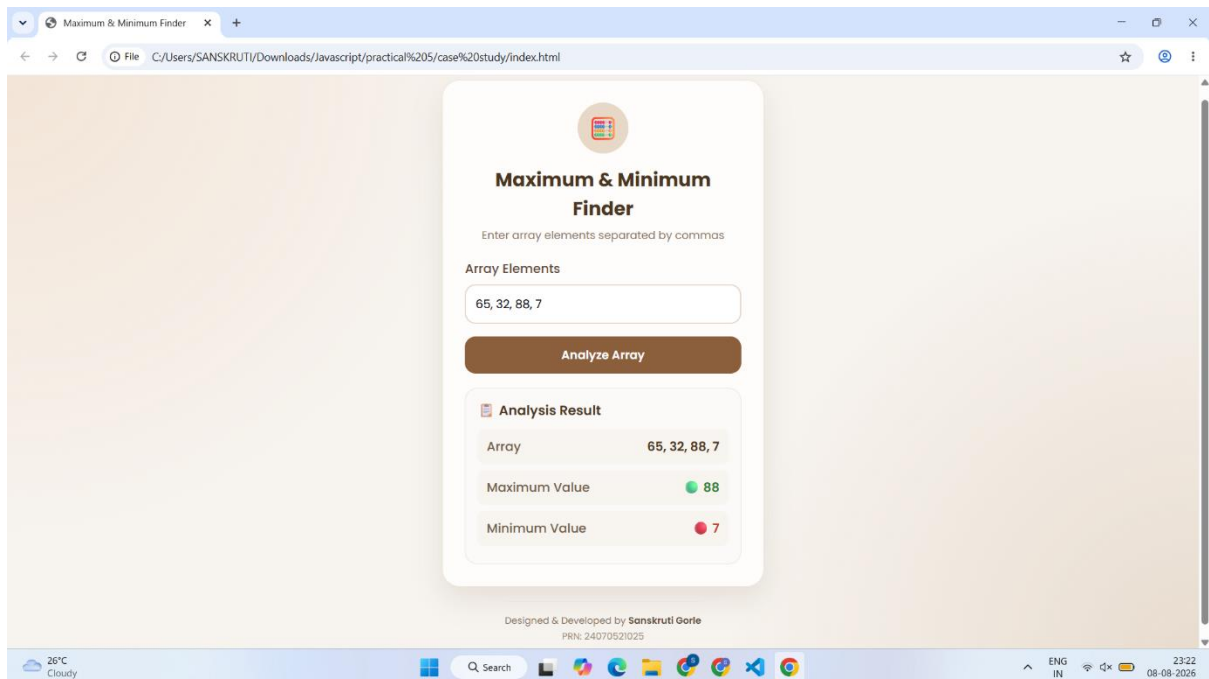
<div class="row">
  <span class="label">Maximum Value</span>
  <span class="value max">□ ${max}</span>
</div>

<div class="row">
  <span class="label">Minimum Value</span>
  <span class="value min">● ${min}</span>
</div>

`;
}

```

Output –



Conclusion –

The experiment was successfully performed to study JavaScript arrays, array manipulation methods, array iteration methods, loops and object handling. A Shopping Cart Calculator was developed to store product objects, calculate the total amount and apply discount logic. A Maximum and Minimum Finder was also implemented using array traversal to determine the largest and smallest values from an array of numbers.